

# Chapter 5 Transform Analysis of LTI Systems

- Frequency response of LTI systems
- Linear constant-coefficient difference equations
- Magnitude and phase
- Minimum phase
- Linear phase
- An LTI system can completely be characterized in the time domain by its *impulse response*. (Note: assuming no initial conditions)

Time:  $y[n] = x[n] * h[n]$

z-transform:  $Y(z) = X(z) \cdot H(z)$ ;  $H(z)$  – System function

Frequency response:  $Y(e^{j\omega}) = X(e^{j\omega}) \cdot H(e^{j\omega})$  ;

$H(e^{j\omega})$  -- Frequency response;

Magnitude or gain:  $|H(e^{j\omega})|$  ,

Phase response or phase shift:  $\angle H(e^{j\omega})$

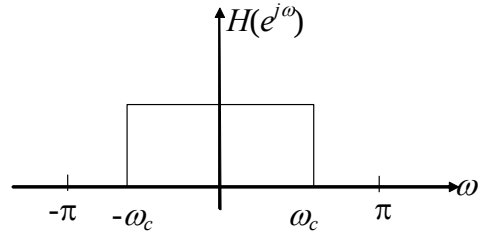
## 5.1 Frequency Response of LTI Systems

- **Frequency-selective filters**

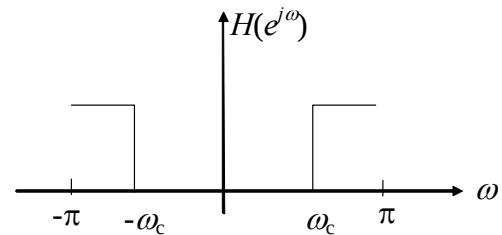
$$H(e^{j\omega}) = \begin{cases} 1, & \omega_1 \leq \omega \leq \omega_2 \\ 0, & \text{otherwise} \end{cases}$$

For example, the ideal low-pass filter is  $\omega_1 = 0$ ,  $\omega_2 = \omega_c$ . The ideal high-pass filter is  $\omega_1 = \omega_c$ ,  $\omega_2 = \pi$ . This type of filters allow only the input signal components in the “1” band to pass through.

Ideal lowpass:  $h_{lp}[n] = \frac{\sin(\omega_c n)}{\pi n}$



Ideal highpass:  $h_{hp}[n] = \delta[n] - \frac{\sin(\omega_c n)}{\pi n}$



*Remark:* This definition includes the phase specification. That is, zero phase for all frequencies. Practical physical systems cannot achieve this specification. In addition, it is non-causal and needs infinite input samples to compute the current output. They are not *computationally realizable*. Typically, the frequency-selective filters are referred to their *magnitude* specifications.

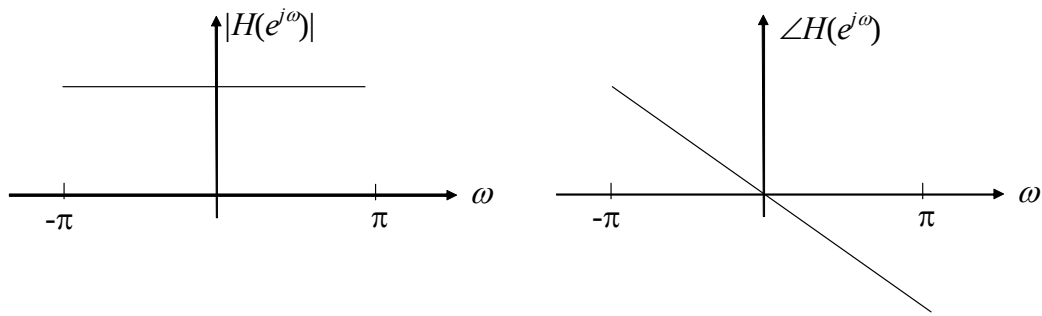
*Terms:* lowpass, highpass, bandpass, bandstop, all-pass filters

- **Phase delay**

$$h_{id}[n] = \delta[n - n_d] \leftrightarrow H_{id}(e^{j\omega}) = e^{-j\omega n_d}$$

The output is a delayed version of the input. (The shape of the waveform is not changed.)

$$\begin{cases} |H_{id}(e^{j\omega})| = 1 \\ \angle H_{id}(e^{j\omega}) = -\omega n_d, \quad |\omega| < \pi \end{cases}$$



**Linear phase:** The phase response is a linear function of  $\omega$  (passing through the origin).

A frequency-selective filter with a linear phase is often acceptable and can be approximated by a practical system. That is,

$$H(e^{j\omega}) = \begin{cases} e^{-j\omega n_d}, & \omega_1 \leq \omega \leq \omega_2 \\ 0, & \text{otherwise} \end{cases}$$

**Group delay:**

$$\tau(\omega) \equiv \text{grd}[H(e^{j\omega})] = -\frac{d}{d\omega} \left\{ \arg[H(e^{j\omega})] \right\}$$

A convenient measure of the linearity of the phase. It is clear that for the ideal delay,  $\tau(\omega) = n_d$  is a constant (independent of  $\omega$ ).

*Motivation:* Narrow-band signals in communication systems

Input:  $x[n] = s[n]\cos(\omega_0 n)$

System:  $\angle H(e^{j\omega}) \approx \begin{cases} -\phi_0 - \omega n_d, & \text{for } \omega > 0 \\ \phi_0 - \omega n_d, & \text{for } \omega < 0 \end{cases}$

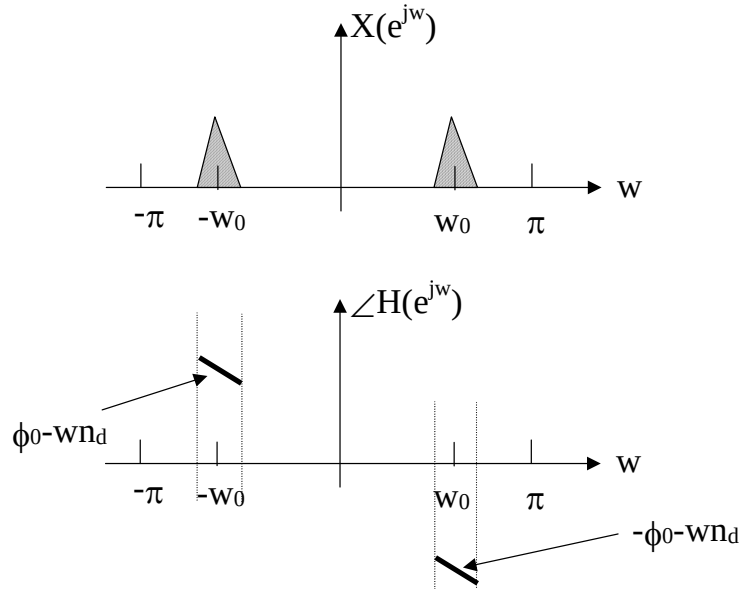
Output:  $y[n] = s[n - n_d]\cos(\omega_0 n - \phi_0 - \omega_0 n_d)$

*Remark:*  $\arg[H(\cdot)]$ : continuous phase ( $|\text{value}| < \pi$  or  $> \pi$ )

$\text{ARG}[H(\cdot)]$ : principal value ( $|\text{value}| < \pi$ )

$$X(e^{j\omega}) = \frac{1}{2}S(e^{j(\omega+\omega_0)}) + \frac{1}{2}S(e^{j(\omega-\omega_0)})$$

$$Y(e^{j\omega}) \approx \frac{1}{2}S(e^{j(\omega+\omega_0)})e^{j(\phi_0-\omega n_d)} + \frac{1}{2}S(e^{j(\omega-\omega_0)})e^{j(-\phi_0-\omega n_d)}$$



## 5.2 Systems Described by Linear Constant-coefficient Difference Equations

$$\begin{aligned} \sum_{k=0}^N a_k y[n-k] &= \sum_{k=0}^M b_k x[n-k] & H(z) &= \frac{Y(z)}{X(z)} \\ \sum_{k=0}^N a_k z^{-k} Y(z) &= \sum_{k=0}^M b_k z^{-k} X(z) & &= \frac{\sum_{k=0}^M b_k z^{-k}}{\sum_{k=0}^N a_k z^{-k}} \\ & & &= \left( \frac{b_0}{a_0} \right) \frac{\prod_{k=1}^M (1 - c_k z^{-1})}{\prod_{k=1}^N (1 - d_k z^{-1})} \end{aligned}$$

*Note:* Such a system is not always a linear system. (Why?)

- **Stability (BIBO)**

BIBO stable iff  $\sum_{n=-\infty}^{\infty} |h[n]| < \infty$  (absolutely summable)

$\Leftrightarrow$  ROC contains the unit circle.

- **Causality**

Causal (right sequence) iff ROC:  $|z| > r_{\max}$

- **Causal and Stable:** All poles are inside the unit circle.

Example:

$$y[n] - \frac{5}{2}y[n-1] + y[n-2] = x[n]$$

$$\Rightarrow H(z) = \frac{1}{(1 - \frac{1}{2}z^{-1})(1 - 2z^{-1})}$$

*Remark:* **Schur-Cohn Stability Test** (P&M, 3<sup>rd</sup> ed., p.213)

To test if all the roots of  $A(z)$  lie inside the unit circle.

$$A(z) = 1 + a_1 z^{-1} + a_2 z^{-2} + \dots + a_N z^{-N}$$

$$\text{Let } A_m(z) = 1 + a_m(1)z^{-1} + a_m(2)z^{-2} + \dots + a_m(m)z^{-m}$$

Define

$$B_m(z) = z^{-m} A_m(z^{-1}) = z^{-m} + a_m(1)z^{-(m-1)} + a_m(2)z^{-(m-2)} + \dots + a_m(m)$$

$$\text{Set } A_N(z) = A(z), \quad K_N = a_N(N)$$

$$\Rightarrow A_{m-1}(z) = \frac{A_m(z) - K_m B_m(z)}{1 - K_m^2} \quad m = N, N-1, \dots, 1$$

where  $K_m = a_m(m)$

If  $|K_m| < 1$  for  $m = 1, 2, \dots, N$

$\Leftrightarrow$  all the roots of  $A(z)$  lie inside the unit circle.

Example:

$$H(z) = \frac{1}{1 - \frac{7}{4}z^{-1} - \frac{1}{2}z^{-2}}$$

$$A_2(z) = 1 - \frac{7}{4}z^{-1} - \frac{1}{2}z^{-2} \quad K_2 = -\frac{1}{2}$$

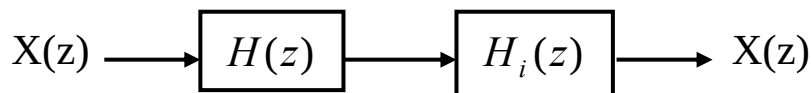
$$\Rightarrow B_2(z) = -\frac{1}{2} - \frac{7}{4}z^{-1} + z^{-2}$$

$$\Rightarrow A_1(z) = \frac{A_2(z) - K_2 B_2(z)}{1 - K_2^2} = 1 - \frac{7}{2}z^{-1}$$

$$\Rightarrow K_1 = -\frac{7}{2}$$

$\Rightarrow$  unstable

- **Inverse System:**



$$H(z)H_i(z) = 1$$

$$\text{Or, } H(z) = \frac{1}{H_i(z)}$$

Poles of  $H(z) \rightarrow$  Zeros of  $H_i(z)$

Zeros of  $H(z) \rightarrow$  Poles of  $H_i(z)$

$H_i(z)$  is *causal and stable* if all the zeros of  $H(z)$  are inside the unit circle.

**Minimum-phase:** If both poles and zeros are inside the unit circle.

- **Rational System Functions**

**IIR (Infinite Impulse Response):**

$$H(z) = \sum_{r=0}^{M-N} B_r z^{-r} + \sum_{k=1}^N \frac{A_k}{1 - d_k z^{-1}}$$

At least one nonzero pole  $\{d_k\}$  is not cancelled by zero. The impulse

response contains terms like  $h[n] = \sum_{k=1}^N A_k (d_k)^n$ .

**FIR (Finite Impulse Response):**

$$H(z) = \sum_{k=0}^M b_k z^{-k}$$

No nonzero poles (All poles are at “0”).

*Ex.*

$$h[n] = \begin{cases} a^n, & 0 \leq n \leq M \\ 0, & \text{otherwise} \end{cases}$$

## 5.3 Frequency Response for Rational System Functions

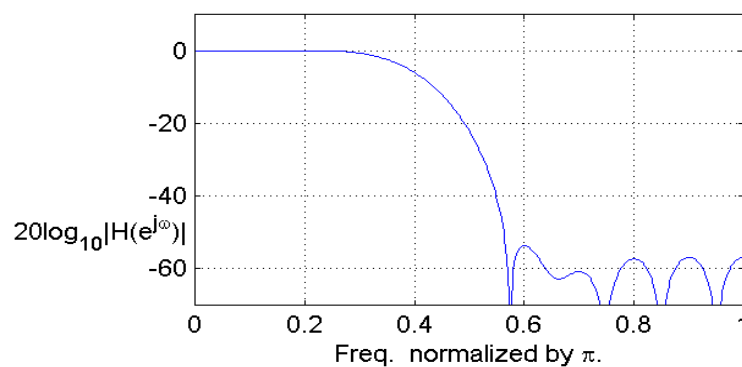
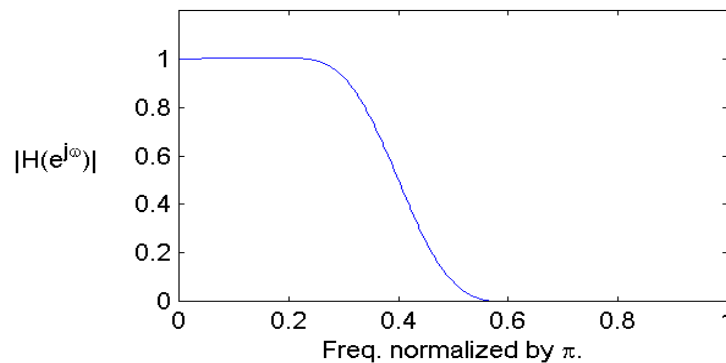
$$H(e^{j\omega}) = \frac{\sum_{k=0}^M b_k e^{-j\omega k}}{\sum_{k=0}^N a_k e^{-j\omega k}} = \left( \frac{b_0}{a_0} \right) \frac{\prod_{k=1}^M (1 - c_k e^{-j\omega k})}{\prod_{k=1}^N (1 - d_k e^{-j\omega k})}$$

Log magnitude: unit in *decibels* (dB)

Gain in dB = - *Attenuation* in dB

$$20 \log_{10} |H(e^{j\omega})| = 20 \log_{10} \left| \left( \frac{b_0}{a_0} \right) \right| + \sum_{k=1}^M 20 \log_{10} |1 - c_k e^{-j\omega k}| - \sum_{k=1}^N 20 \log_{10} |1 - d_k e^{-j\omega k}|$$

Clearly,  $20 \log_{10} |Y(e^{j\omega})| = 20 \log_{10} |H(e^{j\omega})| + 20 \log_{10} |X(e^{j\omega})|$





Phase:

$$\angle H(e^{j\omega}) = \angle \left( \frac{b_0}{a_0} \right) + \sum_{k=1}^M \angle(1 - c_k e^{-j\omega k}) - \sum_{k=1}^N \angle(1 - d_k e^{-j\omega k})$$

Group delay:

$$\text{grd}[H(e^{j\omega})] = \sum_{k=1}^N \frac{d}{d\omega} (\arg[1 - d_k e^{-j\omega k}]) - \sum_{k=1}^M \frac{d}{d\omega} (\arg[1 - c_k e^{-j\omega k}])$$

- *Problem? Phase ambiguity.* The real filter phase is continuous and can be greater than  $\pi$  or smaller than  $-\pi$ . However, the calculated phase is the principal value,

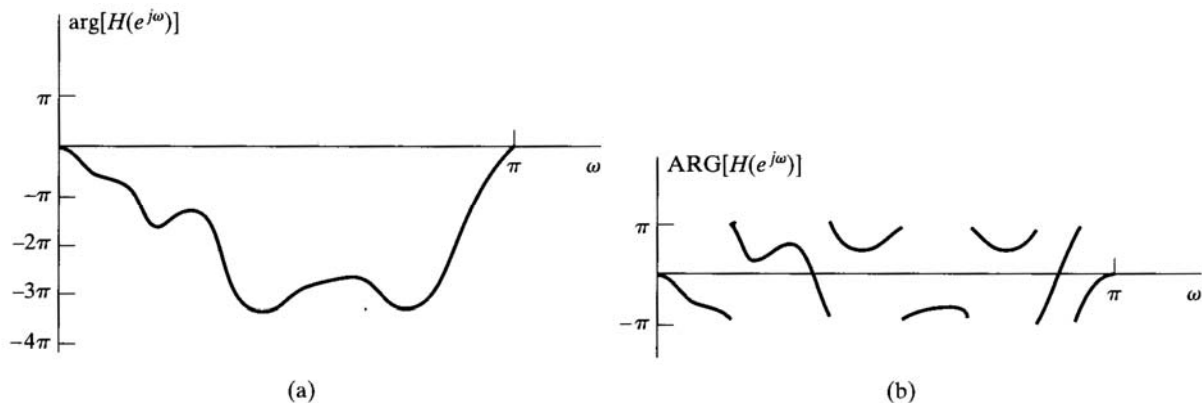
$$-\pi < \text{ARG}[H(e^{j\omega})] < \pi$$

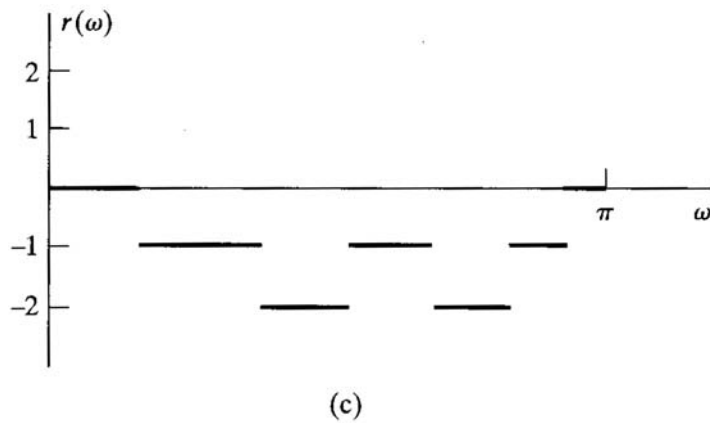
Therefore,  $\angle H(e^{j\omega}) = \text{ARG}[H(e^{j\omega})] + 2\pi \cdot r(\omega)$  where  $r(\omega)$  is an integer and can be a function of  $\omega$ . How to find  $r(\omega)$ ? It can be rather complicated and we skip the details here.

However, the phase *group delay* is identical in both cases (except for the discontinuities of  $\text{ARG}[H(e^{j\omega})]$ ).

$$\frac{d}{d\omega} \{ \text{ARG}[H(e^{j\omega})] \} = \frac{d}{d\omega} \{ \arg[H(e^{j\omega})] \}$$

*Ex. Continuous-phase (Fig. 5.1, p.304)*





**Fig. 5.1** (a) Continuous-phase curve for a system function evaluated on the unit circle. (b) Principal value of the phase curve in part (a). (c) Integer multiples of  $2\pi$  to be added to  $\text{ARG}[H(e^{j\omega})]$  to obtain  $\arg[H(e^{j\omega})]$ .

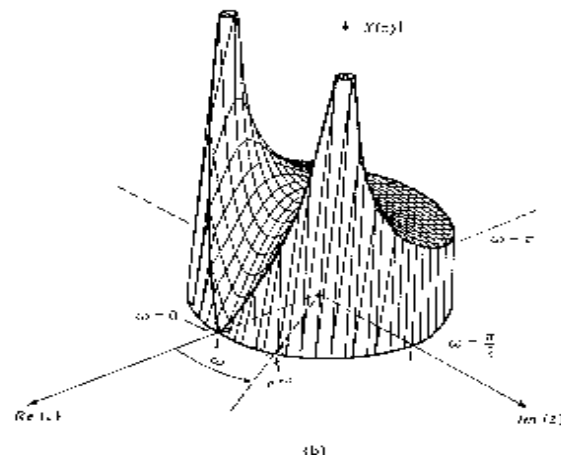
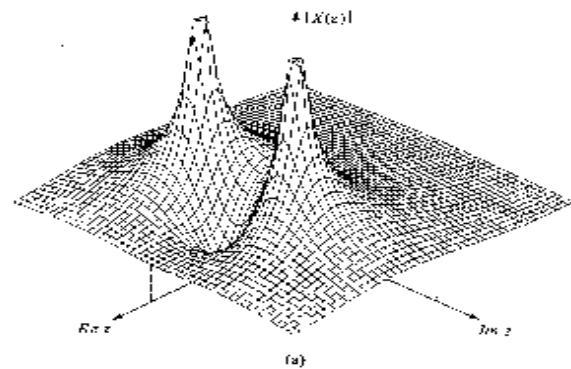
**Example of frequency response** (3-D plot, P&M, 3<sup>rd</sup> ed., p.177)

$$X(z) = \frac{z^{-1}(1 - z^{-1})}{1 - 1.2732z^{-1} + .81z^{-2}}$$

zeros: 1

poles:  $0.9 \cdot e^{\pm j\pi/4}$

$|r| \approx 1$ , resonator



- Resonator, Notch filter, and Comb filter

-- **Resonator**: Narrow resonant peaks (magnitude); eg., complex poles near the unit circle.

$$H(z) = \frac{b_0}{(1 - re^{j\theta}z^{-1})(1 - re^{-j\theta}z^{-1})}$$

-- **Notch Filter**: Narrow deep notches (nulls); e.g., complex zeros

on/near the unit circle.

$$H(z) = b_0(1 - re^{j\theta}z^{-1})(1 - re^{-j\theta}z^{-1})$$

-- **Comb Filter**: Special notch filter in which nulls occur periodically; e.g., complex zeros uniformly distributed on/near the unit circle.

$$H(z) = \left( \frac{1}{M+1} \right) \frac{(1 - z^{-(M+1)})}{(1 - z^{-1})}$$

- **Single zero or pole**

$$H(z) = \frac{(z - z_0)}{(z - p_0)}, \quad \text{zero: } z_0, \text{ pole: } p_0$$

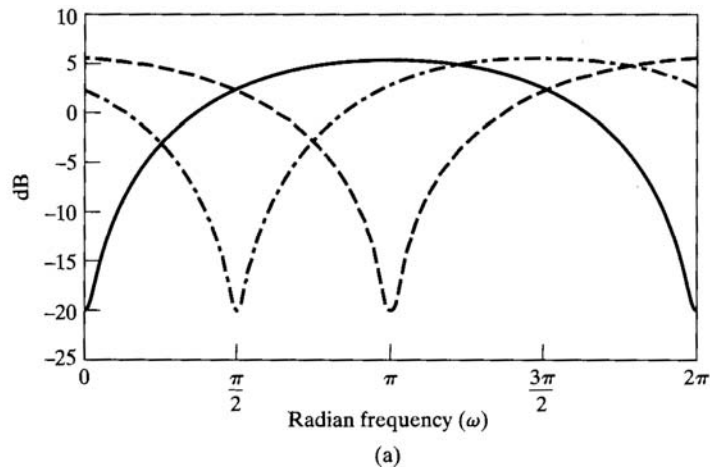
$$\text{Geometric evaluation: } H(e^{j\omega}) = \frac{(e^{j\omega} - z_0)}{(e^{j\omega} - p_0)}$$

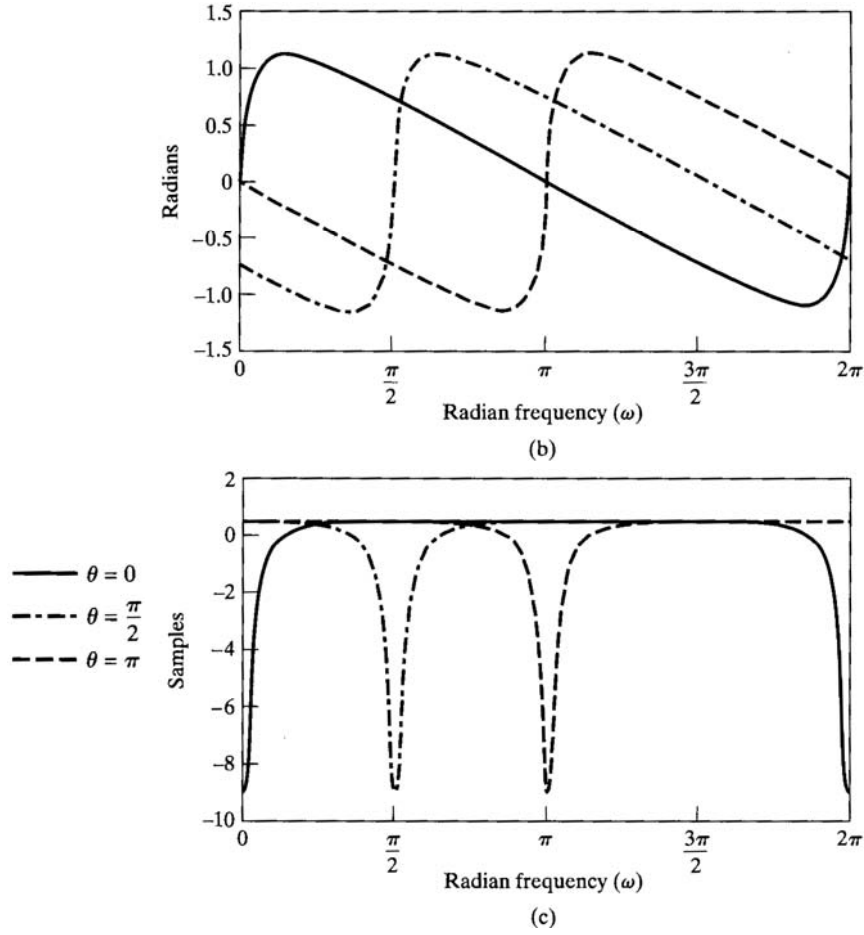
$$\begin{cases} |H(e^{j\omega})| = \frac{|\overrightarrow{e^{j\omega} - z_0}|}{|\overrightarrow{e^{j\omega} - p_0}|} = \frac{|\vec{v}_3|}{|\vec{v}_1|} \\ \angle H(e^{j\omega}) = \angle(e^{j\omega} - z_0) - \angle(e^{j\omega} - p_0) = \theta_{z_0} - \theta_{p_0} \end{cases}$$

$$\text{Factor: } (1 - re^{j\theta}z^{-1})$$

E.g.,  $r = 0.9$ ,  $\theta = 0, \pi/2, \pi$  (Fig. 5.9; p.321)

E.g.,  $r = 1, 0.9, 0.7$ ,  $\theta = \pi$  (Fig. 5.11; p.262)





**Figure 5.9** Frequency response for a single zero, with  $r = 0.9$  and the three values of  $\theta$  shown. (a) Log magnitude. (b) Phase. (c) Group delay.

In general,

$$H(z) = \left( \frac{b_0}{a_0} \right) \frac{\prod_{k=1}^M (1 - c_k z^{-1})}{\prod_{k=1}^N (1 - d_k z^{-1})} = \left( \frac{b_0}{a_0} \right) z^{N-M} \frac{\prod_{k=1}^M (z - c_k)}{\prod_{k=1}^N (z - d_k)}$$

$$\left\{ \begin{array}{l} |H(e^{j\omega})| = \left| \frac{b_0}{a_0} \right| \frac{\prod_{k=1}^M |e^{j\omega} - c_k|}{\prod_{k=1}^N |e^{j\omega} - d_k|} \\ \angle H(e^{j\omega}) = \angle \left( \frac{b_0}{a_0} \right) + \sum_{k=1}^M \angle(e^{j\omega} - c_k) - \sum_{k=1}^N \angle(e^{j\omega} - d_k) + (N - M)\omega \end{array} \right.$$

- **A second-order IIR system** (conjugate poles)

$$H(z) = \frac{1}{(1 - re^{j\theta} z^{-1})(1 - re^{-j\theta} z^{-1})}$$

$$\begin{cases} |H(e^{j\omega})| = \frac{1}{|\vec{v}_1| |\vec{v}_2|} \\ \angle H(e^{j\omega}) = -\angle(e^{j\omega} - re^{j\theta}) - \angle(e^{j\omega} - re^{-j\theta}) \end{cases}$$

E.g.,  $r = 0.9$ ,  $\theta = \pi/4$  (Fig. 5.13; p.325)

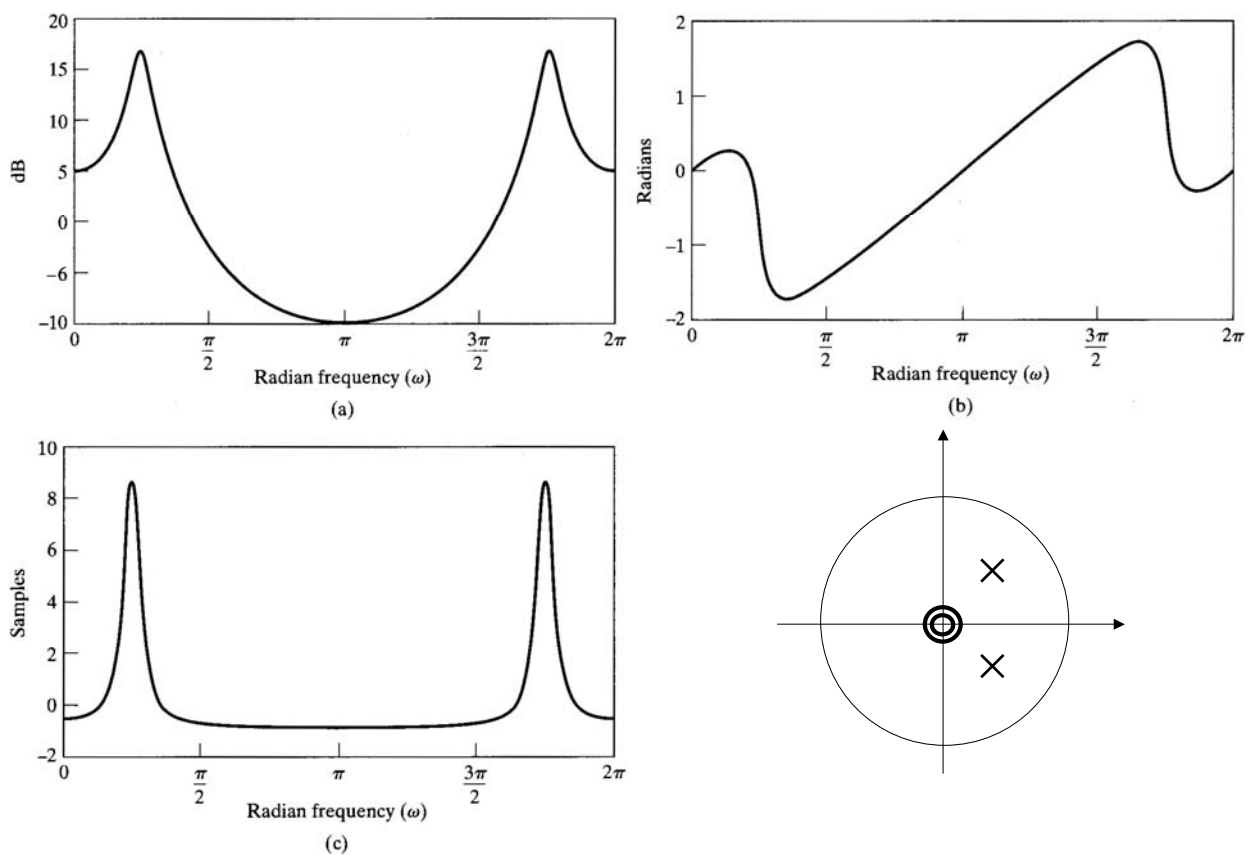


Figure 5.13 Frequency response for a complex-conjugate pair of poles as in Example 5.6, with  $r = 0.9$ ,  $\theta = \pi/4$ . (b) Phase. (c) Group delay.

- **A second-order FIR system** (conjugate zeros)

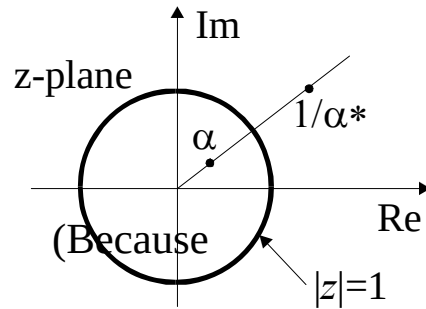
$$H(z) = (1 - re^{j\theta} z^{-1})(1 - re^{-j\theta} z^{-1}) = 1 - 2r \cos \theta z^{-1} + r^2 z^{-2}$$

## 5.4 Relationship Between Magnitude and Phase

- For rational system functions, there is some constraint between magnitude and phase.
- Given the *number* of poles and zeros and the *magnitude (phase) response*, there are only a *finite* number of possible *phase (magnitude) responses*.
- Example: Given magnitude (square), try to decide its phase.

$$\begin{aligned} |H(e^{j\omega})|^2 &= H(e^{j\omega})H^*(e^{j\omega}) \\ &= H(z)H^*\left(\frac{1}{z^*}\right)\bigg|_{z=e^{j\omega}} \end{aligned}$$

$$\begin{aligned} H(z) &= \left(\frac{b_0}{a_0}\right) \frac{\prod_{k=1}^M (1 - c_k z^{-1})}{\prod_{k=1}^N (1 - d_k z^{-1})} \\ H^*\left(\frac{1}{z^*}\right) &= \left(\frac{b_0}{a_0}\right) \frac{\prod_{k=1}^M (1 - c_k^* z)}{\prod_{k=1}^N (1 - d_k^* z)} \end{aligned}$$



$$(1 - c_k z^{-1})^* = (1 - c_k^* (z^{-1})^*) \Rightarrow (1 - c_k^* z) \text{ for } z \rightarrow (z^*)^{-1}$$

Let

$$C(z) = H(z)H^*\left(\frac{1}{z^*}\right) = \left(\frac{b_0}{a_0}\right)^2 \frac{\prod_{k=1}^M (1 - c_k z^{-1})}{\prod_{k=1}^N (1 - d_k z^{-1})} \frac{\prod_{k=1}^M (1 - c_k^* z)}{\prod_{k=1}^N (1 - d_k^* z)}$$

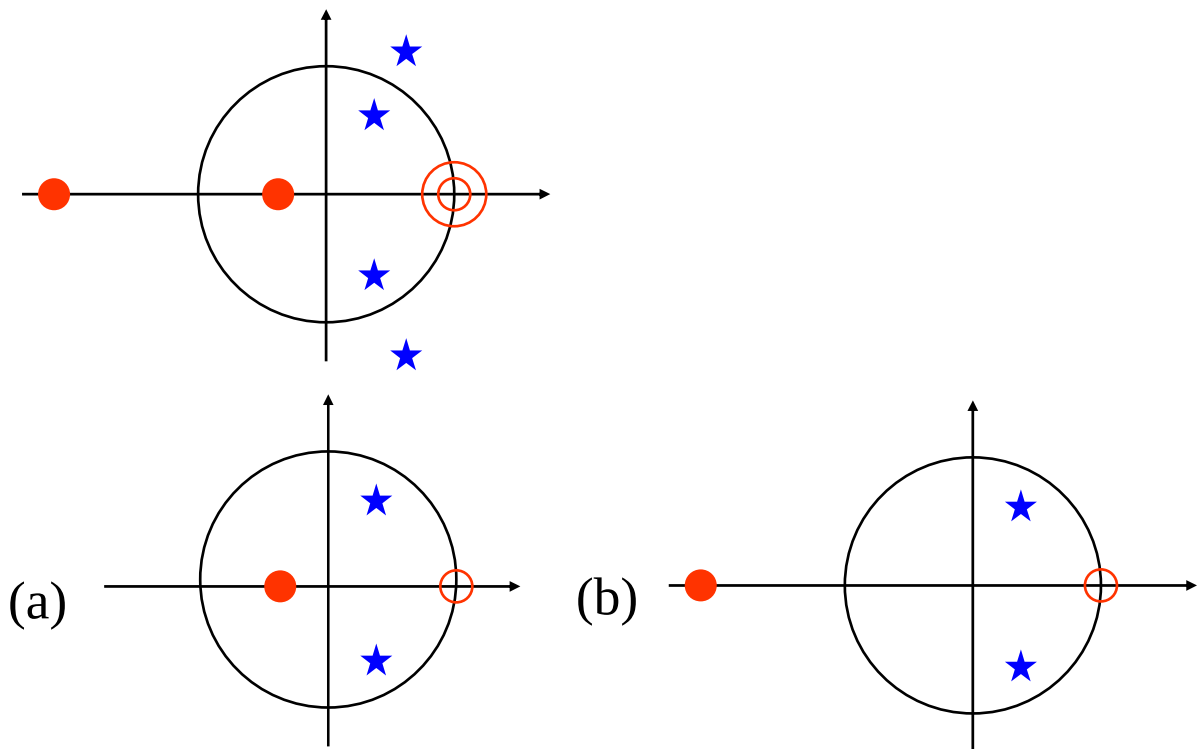
For each pole  $d_k$  of  $H(z) \Rightarrow$  poles  $d_k$  and  $(d_k^*)^{-1}$  of  $C(z)$

For each zero  $c_k$  of  $H(z) \Rightarrow$  zeros  $c_k$  and  $(c_k^*)^{-1}$  of  $C(z)$

If  $H(z)$  is causal and stable, all its poles are inside the unit circle.

Given  $C(z)$ , we may want to find a causal, stable and real  $H(z)$ .

Ex., Fig.5.16, p.331.

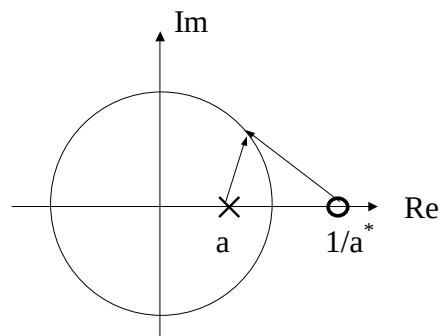


## 5.5 All-pass Systems

- Magnitude = constant; all frequency components can pass through (but the phase is not linear).

$$H_{\text{ap}}(z) = \frac{z^{-1} - a^*}{1 - az^{-1}} \quad \begin{array}{l} \text{pole: } a \\ \text{zero: } \frac{1}{a^*} \end{array}$$

$$\begin{aligned} H_{\text{ap}}(e^{j\omega}) &= \frac{e^{-j\omega} - a^*}{1 - ae^{-j\omega}} \\ &= e^{-j\omega} \cdot \frac{1 - a^* e^{j\omega}}{1 - ae^{-j\omega}} \end{aligned}$$



$$\text{For } a = re^{j\theta}, \quad \begin{cases} |H_{\text{ap}}(e^{j\omega})| = \text{constant} \\ \angle \left[ \frac{e^{-j\omega} - re^{j\theta}}{1 - re^{j\theta}e^{-j\omega}} \right] = -\omega - 2 \arctan \left[ \frac{r \sin(\omega - \theta)}{1 - r \cos(\omega - \theta)} \right] \end{cases}$$

Ex., Figs. 5.19, 5.20, p.335, 336.

Note:  $\theta = 0$  or  $\theta = \pi$ , phase discontinuity (due to  $\arctan()$ ) at  $\omega = \pi$ .

$$H_{ap}(z) = \frac{(z^{-1} - 0.9e^{-j\frac{\pi}{4}})(z^{-1} - 0.9e^{j\frac{\pi}{4}})}{(1 - 0.9e^{j\frac{\pi}{4}}z^{-1})(1 - 0.9e^{-j\frac{\pi}{4}}z^{-1})}$$

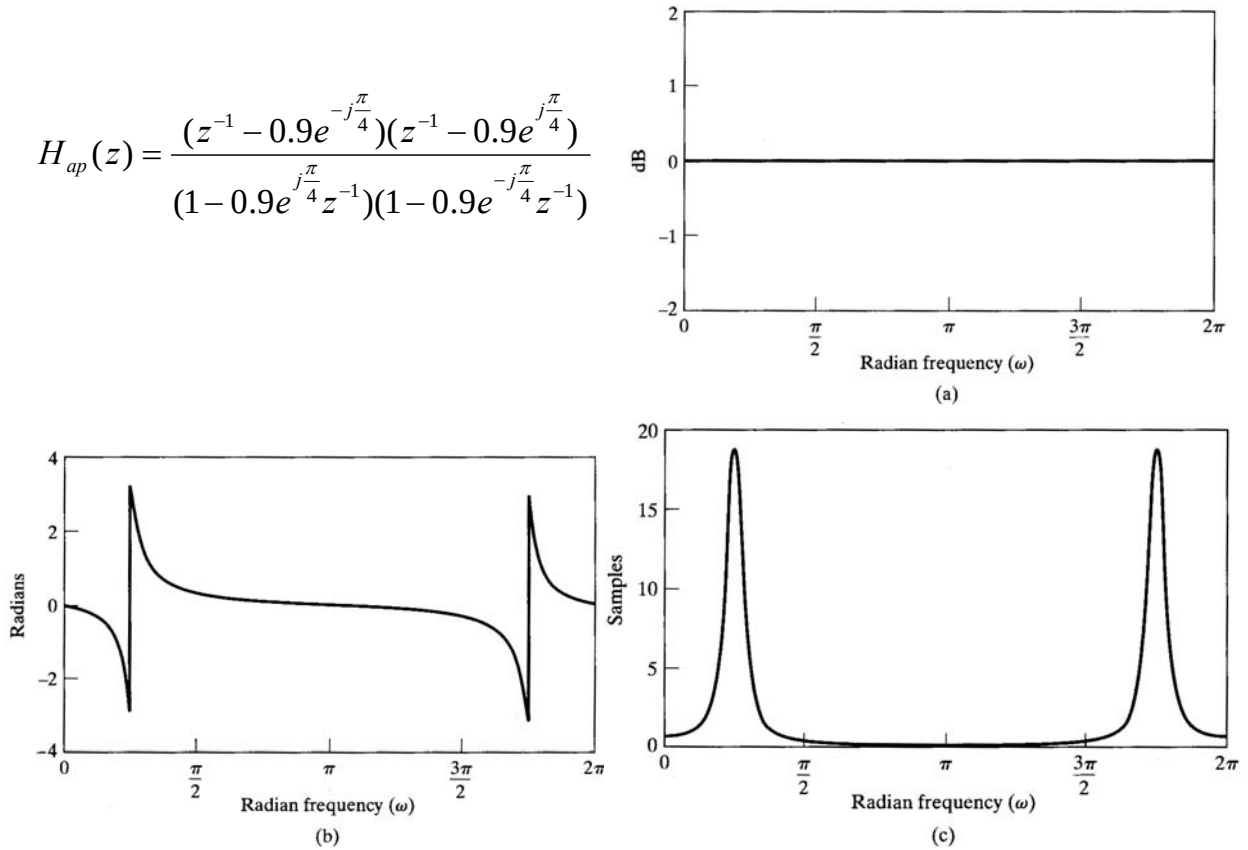


Figure 5.20 Frequency response of 2<sup>nd</sup>-order all-pass system with poles at  $z = 0.9e^{\pm j\pi/4}$ . (a) Log magnitude. (b) Phase (principal value). (c) Group delay.

- A general all-pass filter is a product of the first-order and second-order factors.
- The phase of a causal all-pass filter is non-positive. It can be shown that the group delay of a first-order all-pass system is non-negative.

$$(Pf) \text{ For } a = re^{j\theta}, \quad \text{grd} \left[ \frac{e^{-j\omega} - re^{-j\theta}}{1 - re^{j\theta}e^{-j\omega}} \right] = \frac{1 - r^2}{1 + r^2 - 2r \cos(\omega - \theta)} = \frac{1 - r^2}{|1 - re^{j\theta}e^{-j\omega}|^2} \geq 0$$

$$\arg[H_{ap}(e^{j\omega})] = -\int_0^\omega \text{grd}[H_{ap}(e^{j\phi})]d\phi + \arg[H_{ap}(e^{j0})] \leq 0, \text{ for } 0 \leq \omega < \pi$$



(Details of proof are on p.337.)

- All-pass filters can be used as phase compensators. They are useful in transforming frequency-selective low-pass filters into other frequency-selective form and in obtaining variable-cutoff filters.

## 5.6 Minimum-phase Systems

- *Minimum-phase system:*

$H(z)$  and its inverse  $1/H(z)$  are both causal and stable.

→ All poles and zeros are inside the unit circle.

- Given the specification on magnitude square, determines a unique minimum-phase system
- For any (stable, causal) rational system function  $H(z)$ , it can be expressed by

$$H(z) = H_{\min}(z) \cdot H_{ap}(z)$$

(Pf) Suppose  $H(z)$  has one zero ( $z = 1/c^*$ ,  $|c| < 1$ ) outside the unit circle (and the remaining poles and zeros are inside the unit circle).

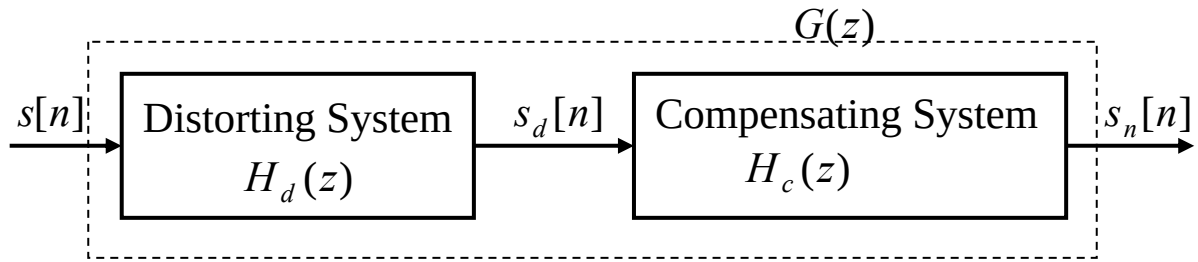
Then,

$$\begin{aligned} H(z) &= H_1(z)(z^{-1} - c^*) \\ &= H_1(z)(1 - cz^{-1}) \frac{z^{-1} - c^*}{1 - cz^{-1}} \end{aligned}$$

Note:  $H_1(z)(1 - cz^{-1})$  is minimum phase.

The above procedure can be extended to general cases.

- A use of minimum-phase system – frequency-response compensation



Design  $H_c(z)$  such that  $G(z) = H_d(z) \cdot H_c(z)$  is desired.

For example, if we wish  $G(z) = H_d(z) \cdot H_c(z) = \text{const}$

$$\text{Let } H_d(z) = H_{d\min}(z) \cdot H_{ap}(z) \text{ then } H_c(z) = \frac{1}{H_{d\min}(z)}$$

Ex., p.342; FIR filter

4 zeros:  $0.9e^{\pm j0.6\pi}$  inside the unit circle,  
 $1.25e^{\pm j0.8\pi}$  outside the unit circle.

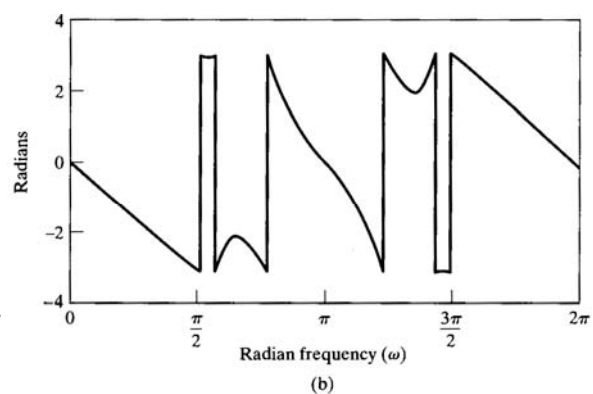
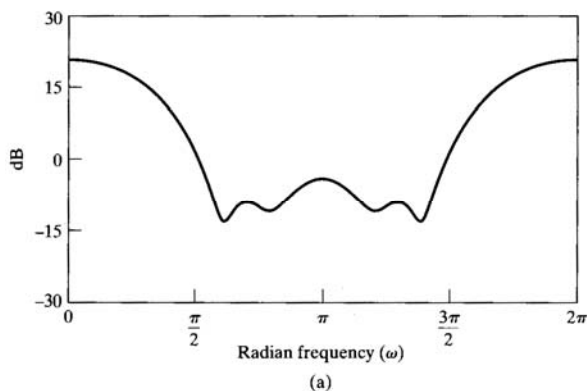
$$H(z)$$

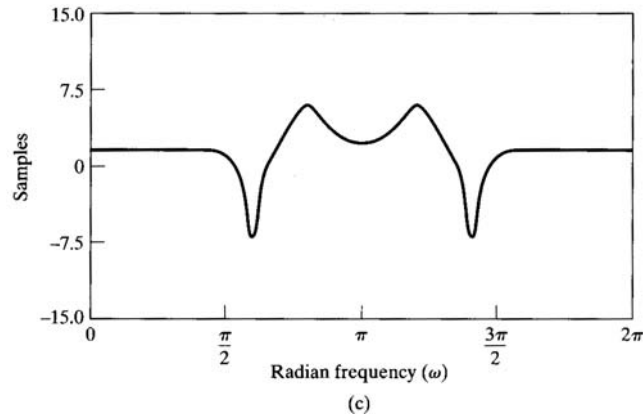
$$H_{\min}(z) \leftrightarrow H_c(z) = \frac{1}{H_{\min}(z)}$$

$$\begin{aligned} H_{\text{overall}}(z) &= H(z) \cdot H_c(z) \\ &= H_{ap}(z) \end{aligned}$$

Example:

$$H(z) = (1 - 0.9e^{j0.6\pi} z^{-1})(1 - 0.9e^{-j0.6\pi} z^{-1})(1 - 1.25e^{j0.8\pi} z^{-1})(1 - 1.25e^{-j0.8\pi} z^{-1})$$



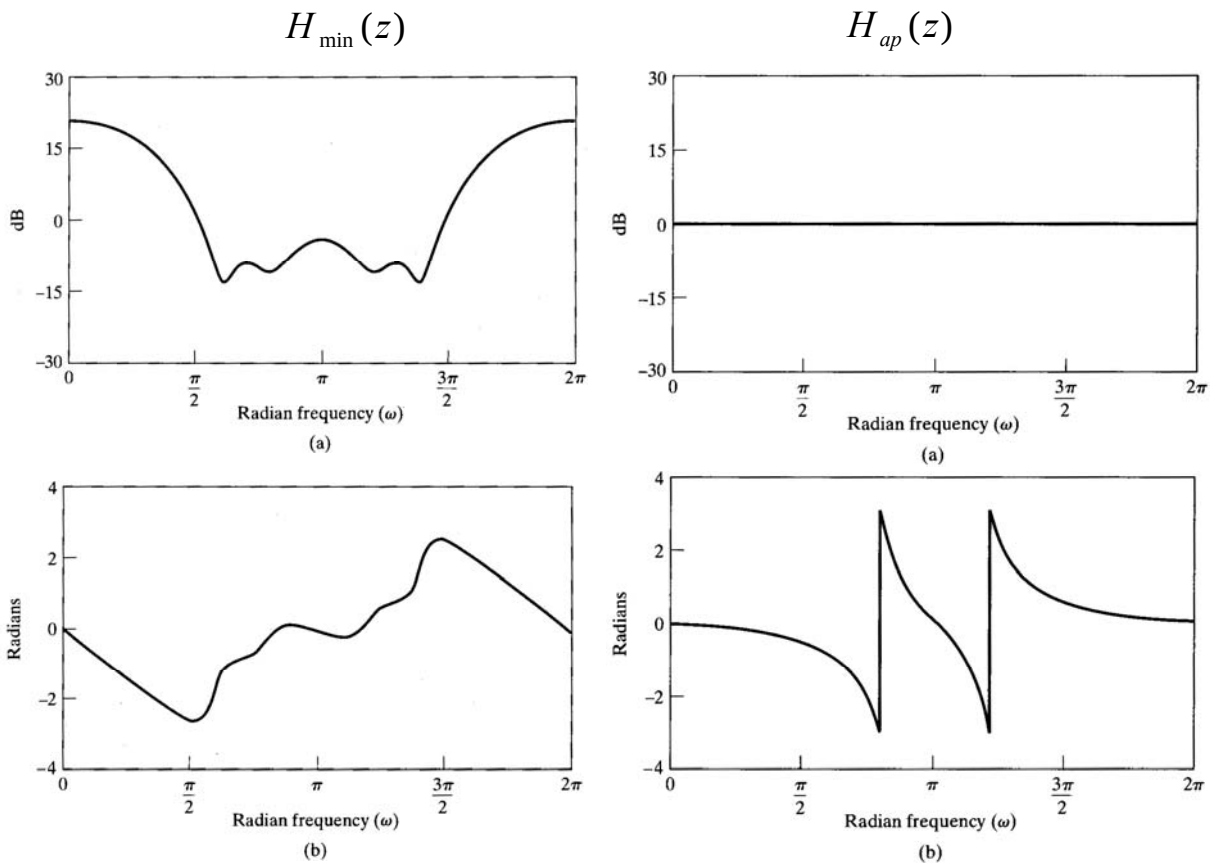


**Figure 5.24** Frequency response for FIR system with pole–zero plot in Figure 5.23. (a) Log magnitude. (b) Phase (principal value). (c) Group delay.

$$H(z) = H_{\min}(z)H_{ap}(z)$$

$$H_{\min}(z) = (1.25)^2 (1 - 0.9e^{j0.6\pi} z^{-1})(1 - 0.9e^{-j0.6\pi} z^{-1})(1 - 0.8e^{j0.8\pi} z^{-1})(1 - 0.8e^{-j0.8\pi} z^{-1})$$

$$H_{ap}(z) = \frac{(z^{-1} - 0.8e^{-j0.8\pi})(z^{-1} - 0.8e^{j0.8\pi})}{(1 - 0.8e^{-j0.8\pi} z^{-1})(1 - 0.8e^{j0.8\pi} z^{-1})}$$



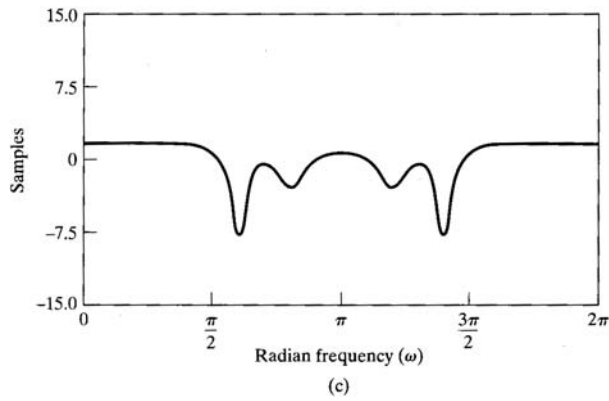


Fig. 5.25

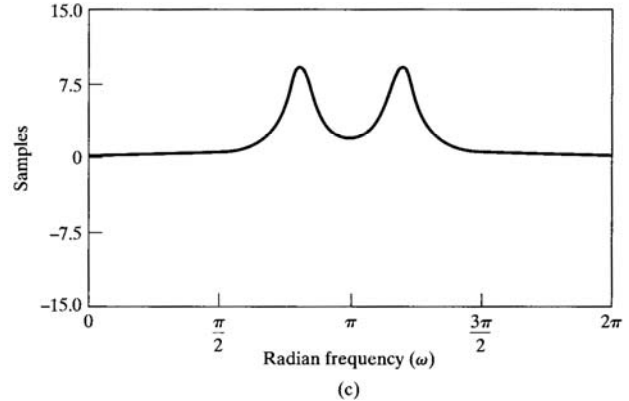


Fig. 5.26

- **Properties**

- **Minimum Phase-lag Property**

Why this type of systems is called “minimum phase”?

Given the magnitude specification  $|H(e^{j\omega})|$  of a system, find the one that has the least phase-lag. ➔ Minimum-phase system.

$$(Pf) \ H(z) = H_{d\min}(z) \cdot H_{ap}(z)$$

$$\arg[H(e^{j\omega})] = \arg[H_{\min}(e^{j\omega})] + \arg[H_{ap}(e^{j\omega})]$$

It was shown that for allpass filters,  $\arg[H_{ap}(e^{j\omega})] < 0$ ,  $0 \leq \omega < \pi$ .

Thus,  $\arg[H(e^{j\omega})] = \arg[H_{\min}(e^{j\omega})] + \text{negative value}$

Hence,  $H_{\min}(z)$  has the minimum phase-lag (less negative).

*Remark:* To ensure the minimum phase-lag property, (in addition to the pole and zero locations),

$$H(e^{j0}) = \sum_{n=-\infty}^{\infty} h[n] > 0$$

- **Minimum Group-delay Property**

$$\text{grd}[H(e^{j\omega})] = \text{grd}[H_{\min}(e^{j\omega})] + \underbrace{\text{grd}[H_{\text{ap}}(e^{j\omega})]}_{>0} \quad 0 \leq \omega \leq \pi$$

$$\Rightarrow \text{grd}[H_{\min}(e^{j\omega})] < \text{grd}[H(e^{j\omega})]$$

### ■ Minimum Energy-delay Property

Same  $|H(e^{j\omega})| \rightarrow$  Different  $H(z)$  (differ in phase)

$(H(z) \leftrightarrow h[n])$

Define **partial energy** (of impulse response) to be

$$E[n] = \sum_{m=0}^n |h[m]|^2.$$

Then the minimum-phase system  $H_{\min}(z)$  has the largest  $E[n]$  among all possible  $H(z)$ . That is, it accumulates more energy up to  $n$ .

*Remarks:* (1) Special case: for  $n=0$ ,  $|h[0]| < |h_{\min}[0]|$ .

(2) Total energy is the same for all systems with the same magnitude response (Parseval's theorem).

(3) **Maximum-phase** system: All its zeros are outside the unit circle.

$\rightarrow$  Maximum energy-delay.

*Ex.,* Four systems with the same  $|H(e^{j\omega})|$ . (p.348)

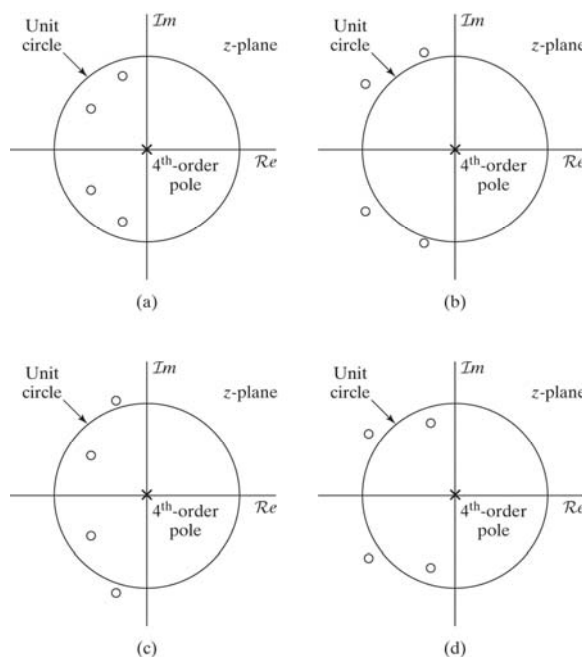
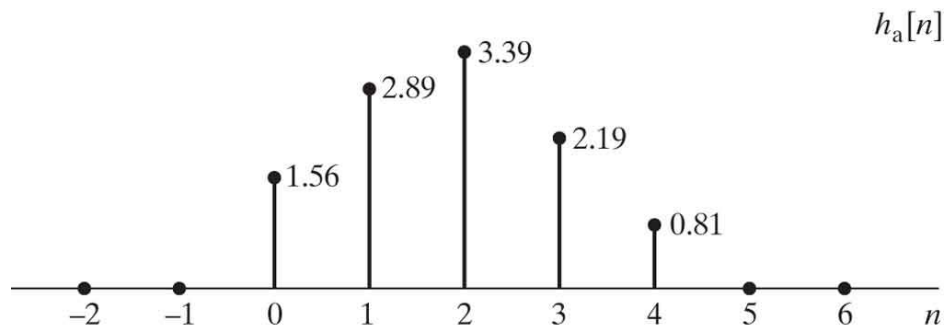
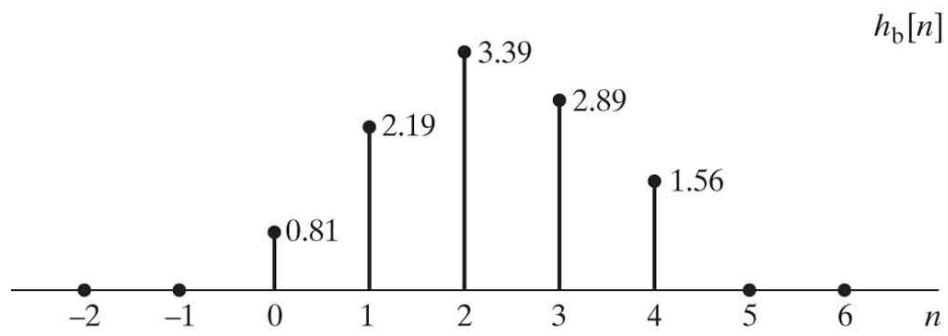


Fig. 5.27



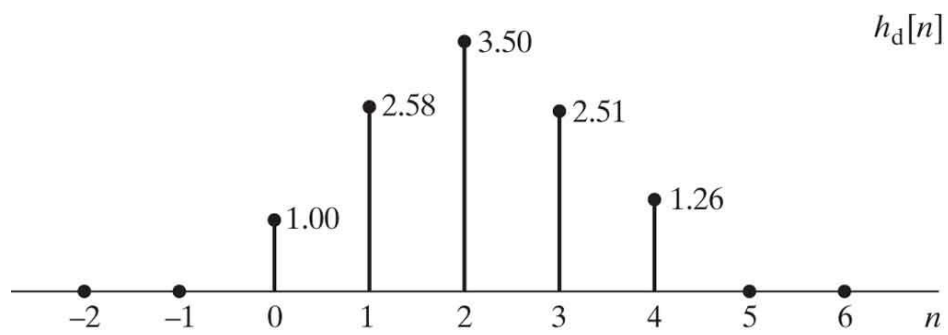
(a)



(b)

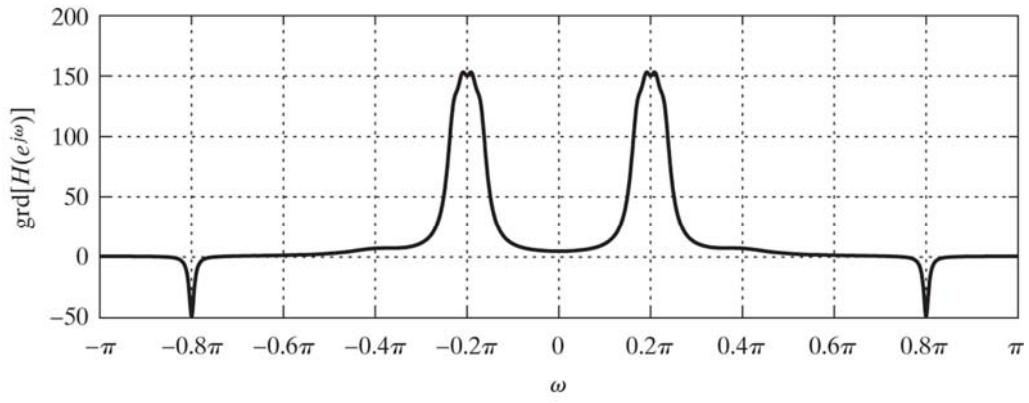


(c)

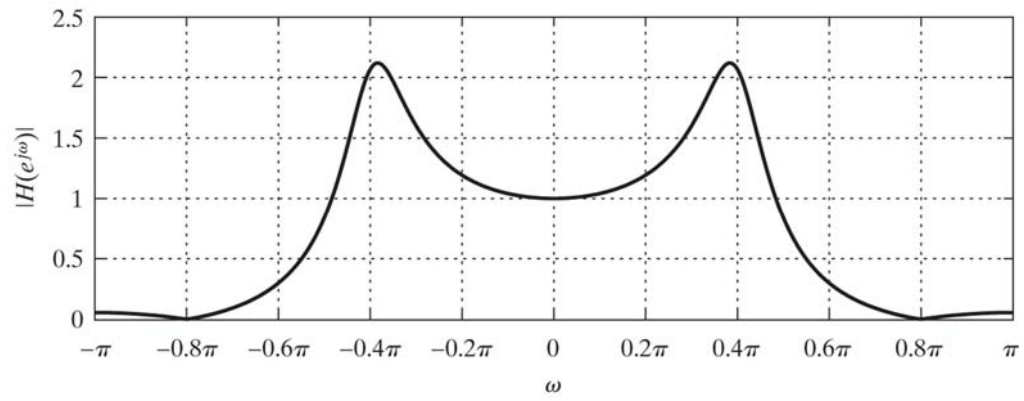


(d)

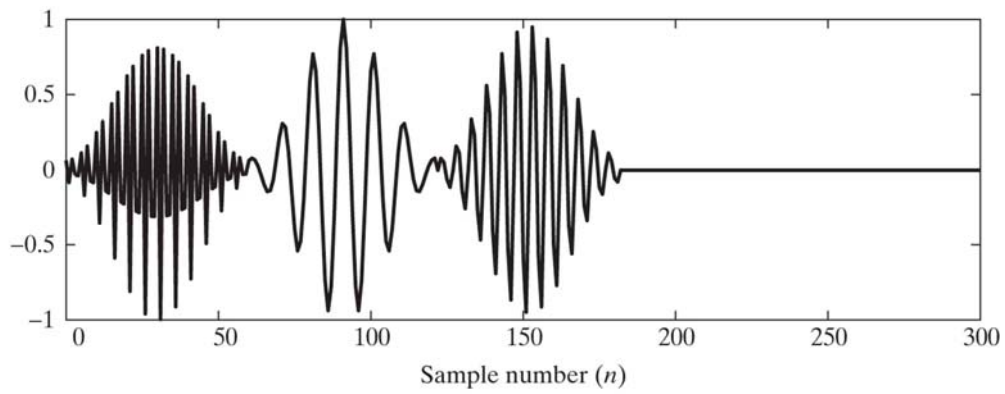
Figure 5.28 Sequences corresponding to the pole-zero plots of Figure 5.27.



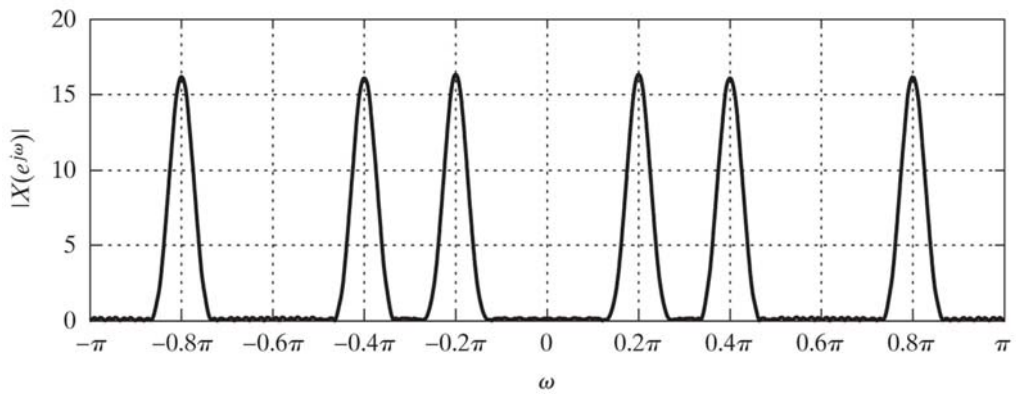
(a) Group delay of  $H(z)$



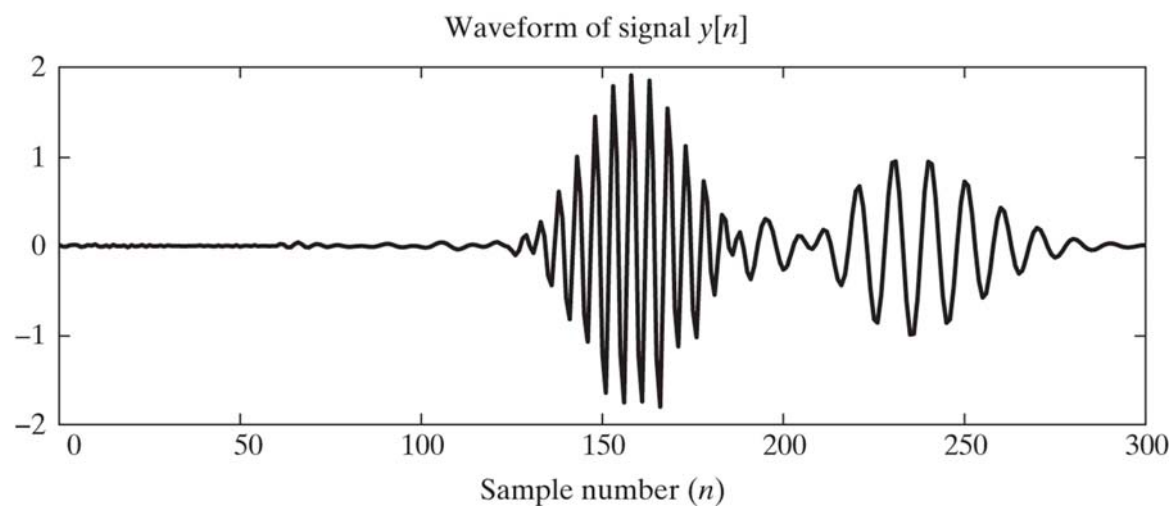
(b) Magnitude of Frequency Response



(a) Waveform of signal  $x[n]$



(b) Magnitude of DTFT of  $x[n]$





## 5.7 Generalized Linear Phase Systems

■ Ideal Delay: 
$$\begin{cases} H_{\text{id}}(e^{j\omega}) = e^{-j\omega\alpha} \\ |H_{\text{id}}(e^{j\omega})| = 1 \\ \angle H_{\text{id}}(e^{j\omega}) = -\omega\alpha \end{cases} \leftrightarrow \begin{cases} h_{\text{id}}[n] = \delta[n - \alpha] \\ \text{grd}[H_{\text{id}}(e^{j\omega})] = \alpha \end{cases}$$

■ Linear phase 
$$\begin{cases} |H(e^{j\omega})| = \text{any} \\ \angle H(e^{j\omega}) = -\omega\alpha, \quad \alpha : \text{can be non - integer} \end{cases}$$

- Why linear phase? (Prof. Lin, Notes 2000)

Barbara after allpass filter



Barbara after lowpass linear phase filter



Ex., Symmetry of  $h[n]$  in the ideal delay system:

$$H_{\text{lp}}(e^{j\omega}) = \begin{cases} e^{-j\omega\alpha}, & |\omega| < \omega_c \\ 0, & \text{otherwise} \end{cases}$$

$$h_{\text{lp}}[n] = \frac{\sin[\omega_c(n - \alpha)]}{\pi(n - \alpha)}$$

1.  $\alpha = \text{integer}$   
 $h_{lp}[n] : \text{even} \quad (h[2\alpha - n] = h[n])$
2.  $\alpha = \text{integer} + \frac{1}{2}$   
 $h_{lp}[n] : \text{even} \quad (h[2\alpha - n] = h[n])$
3. Otherwise,  
no symmetry

We observe a symmetric property,  $h[2\alpha - n] = h[n]$ , in the above linear-phase systems. Is this generally true for *all* linear-phase systems?

### ■ Generalized Linear Phase

$$H(e^{j\omega}) = A(e^{j\omega}) \cdot e^{-j\alpha\omega + j\beta}, \quad \alpha, \beta : \text{real constants}$$

$A(e^{j\omega})$  : real but can be negative

Special case :  $\beta = 0 \rightarrow$  linear phase

$$\begin{cases} \text{Group delay : } \tau(\omega) = \text{grd}[H(e^{j\omega})] = \alpha \\ \text{Phase : } \arg[H(e^{j\omega})] = \beta - \alpha\omega, \quad 0 < \omega < \pi \end{cases}$$

Essentially, this system has a *constant* group delay.

### ■ Symmetry property of a generalized system

$$\begin{aligned} H(e^{j\omega}) &= A(e^{j\omega})e^{j(\beta - \alpha\omega)} \\ &= A(e^{j\omega})\cos(\beta - \alpha\omega) - jA(e^{j\omega})\sin(\beta - \alpha\omega) \end{aligned}$$

On the other hand,

$$H(e^{j\omega}) = \sum_{n=-\infty}^{\infty} h[n]e^{-j\omega n} = \sum_{n=-\infty}^{\infty} h[n]\cos \omega n - j \sum_{n=-\infty}^{\infty} h[n]\sin \omega n$$

$$\text{Tangent of phase} = \frac{\text{Im}}{\text{Re}}$$

$$\tan(\beta - \alpha\omega) = \frac{\sin(\beta - \alpha\omega)}{\cos(\beta - \alpha\omega)} = \frac{-\sum_{n=-\infty}^{\infty} h[n] \sin \omega n}{\sum_{n=-\infty}^{\infty} h[n] \cos \omega n}$$

Note:  $A(e^{j\omega})$  disappear!

Cross-multiplying:

$$\begin{aligned} \sin(\beta - \alpha\omega) \cdot \sum_{n=-\infty}^{\infty} h[n] \cos \omega n &= -\cos(\beta - \alpha\omega) \cdot \sum_{n=-\infty}^{\infty} h[n] \sin \omega n \\ \sum_{n=-\infty}^{\infty} h[n] \{ \cos \omega n \sin(\beta - \alpha\omega) + \sin \omega n \cos(\beta - \alpha\omega) \} &= 0 \\ \sum_{n=-\infty}^{\infty} h[n] \sin(\omega n - \omega\alpha + \beta) &= 0, \quad \forall \omega, \quad \text{eqn. (*)} \end{aligned}$$

This is the necessary conditions of constant group delay. That is,

there is a constraint on the key parameters:  $h[n], \alpha, \beta$ .

There are many solutions that satisfy eqn.(\*).

<Solution 1>

$$\begin{cases} \beta = 0 \text{ or } \pi \\ 2\alpha = M = \text{integer} \\ h[2\alpha - n] = h[n] \end{cases}$$

To see this is a solution, we plug in.

$$\begin{aligned} &\sum_{n=-\infty}^{\infty} h[n] \sin\left(\omega\left(n - \frac{M}{2}\right)\right) \\ &= h\left[\frac{M}{2}\right] \sin\left(\omega\left(\frac{M}{2} - \frac{M}{2}\right)\right) + \sum_{n=\frac{M}{2}+1}^{\infty} h[n] \left( \sin\left(\omega\left(n - \frac{M}{2}\right)\right) + \sin\left(\omega\left(M - n - \frac{M}{2}\right)\right) \right) \\ &= \sum_{n=\frac{M}{2}+1}^{\infty} h[n] \left( \sin\left(\omega\left(n - \frac{M}{2}\right)\right) + \sin\left(\omega\left(\frac{M}{2} - n\right)\right) \right) \\ &= 0 \end{aligned}$$

*Note:* We only require that  $h[n]$  satisfies a symmetric property.

There is no constraint on the numerical values of  $h[n]$ .

*Remark:* It can be shown that  $A(e^{j\omega})$  is even (and real).

<Solution 2>

$$\begin{cases} \beta = \frac{\pi}{2} \text{ or } \frac{3\pi}{2} \\ 2\alpha = M = \text{integer} \\ h[2\alpha - n] = -h[n] \end{cases} \Rightarrow \sin\left(\omega\left(n - \frac{M}{2}\right) + \frac{\pi}{2}\right) = \cos\left(\omega\left(n - \frac{M}{2}\right)\right)$$

Check if this is a valid solution.

$$\begin{aligned} & \sum_{n=-\infty}^{\infty} h[n] \cos\left(\omega\left(n - \frac{M}{2}\right)\right) \\ &= \sum_{n=\frac{M}{2}+1}^{\infty} h[n] \left( \cos\left(\omega\left(n - \frac{M}{2}\right)\right) - \cos\left(\omega\left(\frac{M}{2} - n\right)\right) \right) \\ &= 0 \end{aligned}$$

*Remark:* It can be shown that  $A(e^{j\omega})$  is odd (and real).

There are *other possible* solutions, for example, fractional delay.

### • Causal Generalized Linear-phase Systems

If a generalized linear-phase system is also causal, then eqn.(\*)

becomes

$$\sum_{n=0}^{\infty} h[n] \sin(\omega n - \omega\alpha + \beta) = 0, \quad \forall \omega$$

Under this causal condition and  $h[2\alpha - n] = h[n]$ ,

$$h[n] = 0, \quad n < 0 \text{ and } n > M.$$

*Note:* This is an FIR filter.

More precisely, if the filter length is  $M+1$  and

$$h[n] = \begin{cases} h[M-n], & 0 \leq n \leq M, \\ 0, & \text{otherwise} \end{cases} \text{ (symmetric w.r.t. } M/2)$$

Then,  $H(e^{j\omega}) = A_e(e^{j\omega})e^{-j\omega M/2}$ , where  $A_e(e^{j\omega})$  is real and even.

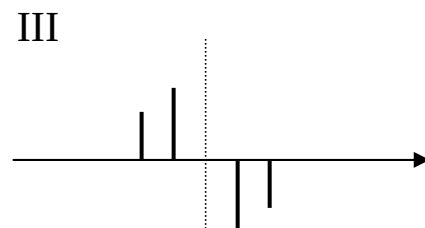
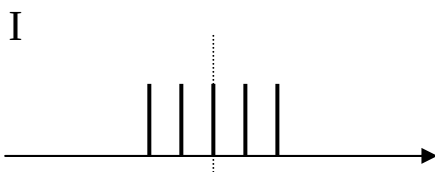
$$\text{If } h[n] = \begin{cases} -h[M-n], & 0 \leq n \leq M, \\ 0, & \text{otherwise} \end{cases} \text{ (anti-symmetric w.r.t. } M/2)$$

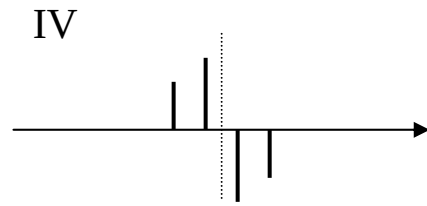
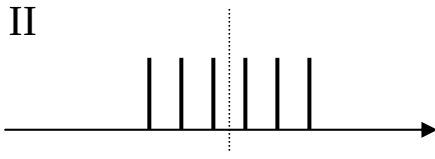
then,  $H(e^{j\omega}) = A_o(e^{j\omega})e^{-j\omega M/2 + j\pi/2}$ , where  $A_o(e^{j\omega})$  is real and odd.

*Remark:* Nearly all linear-phase filters are FIR filters. (There are special types of IIR filters that have linear phase, but they cannot be implemented by difference equations (O&S, p.356).) The above two cases are the most common ones.

#### • Four Types of Linear-phase FIR Filters

$$\left\{ \begin{array}{ll} \text{Type I.} & h[n] = h[M-n] \\ & M \text{ even (or } \frac{M}{2}, \text{ an integer)} \\ \text{Type II.} & h[n] = h[M-n] \\ & M \text{ odd (or } \frac{M}{2}, \text{ a half integer)} \\ \text{Type III.} & h[n] = -h[M-n] \\ & M \text{ even} \\ \text{Type IV.} & h[n] = -h[M-n] \\ & M \text{ odd} \end{array} \right.$$





Example: (Type I)

$$h[n] = \begin{cases} 1 & 0 \leq n \leq 4 \\ 0 & \text{otherwise} \end{cases} \Rightarrow H(e^{j\omega}) = e^{-j2\omega} \frac{\sin(\frac{5\omega}{2})}{\sin(\frac{\omega}{2})}$$

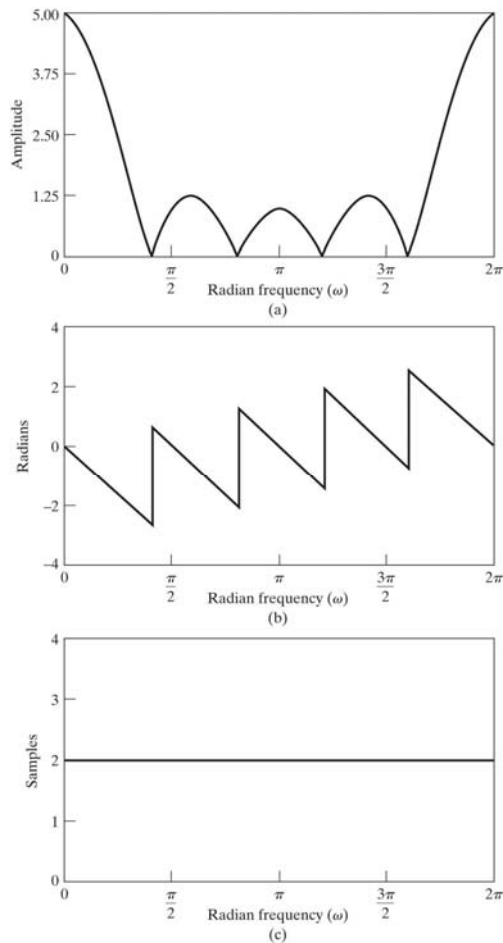


Fig. 5.34

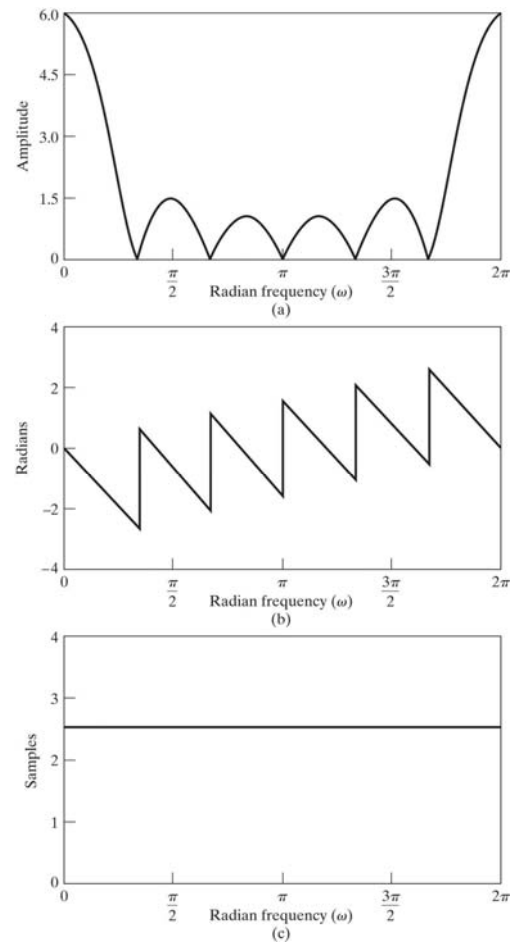


Fig. 5.35

Example: (Type II)

$$h[n] = \begin{cases} 1 & 0 \leq n \leq 5 \\ 0 & \text{otherwise} \end{cases} \Rightarrow H(e^{j\omega}) = e^{-j\frac{5}{2}\omega} \frac{\sin(3\omega)}{\sin(\frac{\omega}{2})}$$

Example: (Type III)

$$h[n] = \delta[n] - \delta[n - 2] \Rightarrow H(e^{j\omega}) = 1 - e^{-j2\omega} = j[2\sin\omega]e^{-j\omega}$$

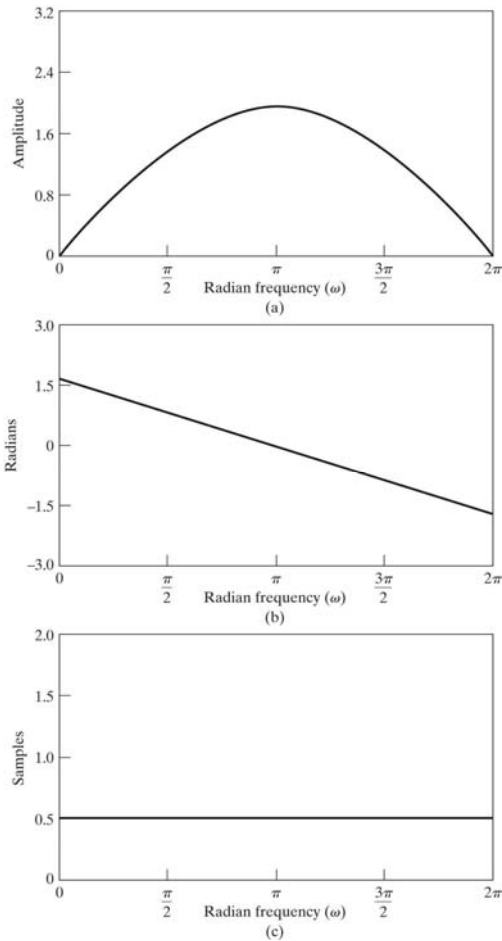


Fig. 5.36

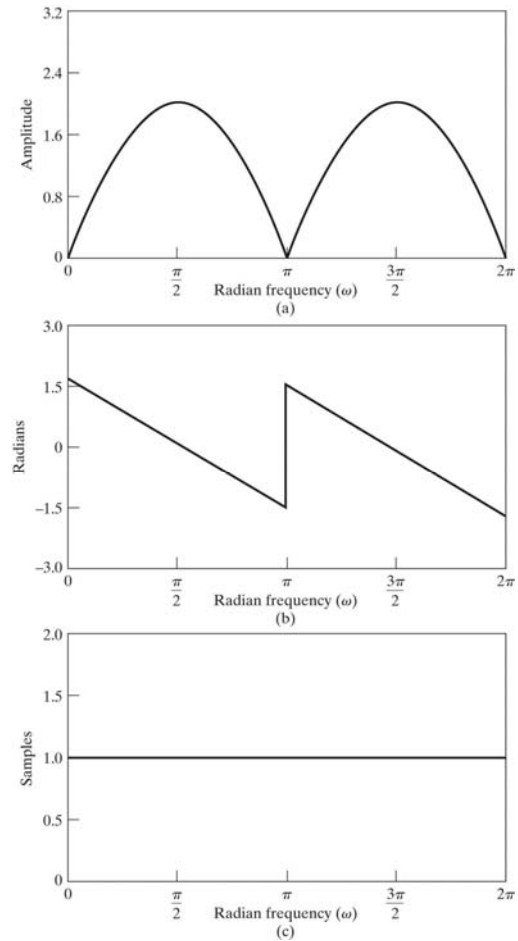


Fig. 5.37

Example: (Type IV)

$$h[n] = \delta[n] - \delta[n - 1] \Rightarrow H(e^{j\omega}) = 1 - e^{-j\omega} = j[2\sin\frac{\omega}{2}]e^{-j\frac{\omega}{2}}$$

- **Zeros of FIR Linear Phase Systems**

- Causal, stable, FIR systems → no finite-value poles
- Zeros are conjugate pairs (because  $h[n]$  is real).
- Zeros are conjugate reciprocals pairs ( $z_0, z_0^{-1}$ ).

(Pf) (A) Type I and II systems

$$\begin{aligned}
 H(z) &= \sum_{n=0}^M h[n] \cdot z^{-n} \\
 &= \sum_{n=0}^M h[M-n] \cdot z^{-n} \\
 &= \sum_{k=M}^0 h[k] \cdot z^k z^{-M} \\
 &= z^{-M} H(z^{-1})
 \end{aligned}$$

(k=M-n)

If  $z_0$  is a zero, i.e.,  $H(z_0) = 0$  then  $z_0^{-M} H(z_0^{-1}) = 0$ .

That is, its reciprocal is a zero too.

(B) Type III and IV systems

$$H(z) = -z^{-M} H(z^{-1})$$

If  $z_0$  is a zero, so does its reciprocal.

- Type II (symmetry, odd) → zero at “-1”

Type III (anti-symmetry, even) → zero at “1” and “-1”

Type IV (anti-symmetry, odd) → zero at “1”

(Pf) (A) Type I and II:

$$H(z) = z^{-M} H(z^{-1})$$

$$\text{Let } z=-1, H(-1) = (-1)^{-M} H(-1)$$

$$H(-1) = (-1)H(-1)$$

When  $M$  is odd,  $H(-1)$  must be 0.



(B) Type III and IV:

$$H(z) = -z^{-M} H(z^{-1})$$

Let  $z=1$ ,  $H(1) = -H(1) \rightarrow H(1)$  must be 0.

Let  $z=-1$ ,  $H(-1) = -(-1)^{-M} H(-1)$ .

When  $M$  is even,  $H(-1) = -H(-1)$  must be 0.

*Remark:* These constraints on the zeros are useful in designing FIR linear-phase systems. For example, for highpass systems,  $M$  cannot be odd (zero should not be at “-1”).

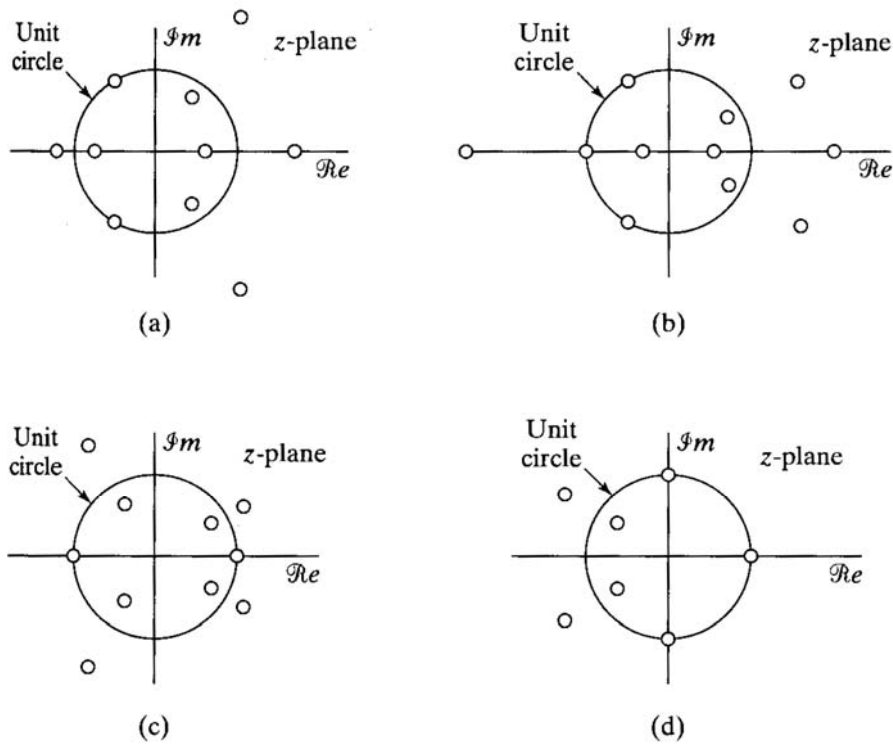


Figure 5.38 Typical plots of zeros for linear-phase systems. (a) Type I. (b) Type II. (c) Type III. (d) Type IV.

■ Any FIR linear-phase system can be expressed as follows:

$$H(z) = \underbrace{H_{\min}(z)}_{\text{linear phase}} \cdot \underbrace{H_{\min}(z)}_{\text{min. phase}} \cdot \underbrace{H_{\text{uc}}(z)}_{\text{unit circle}} \cdot \underbrace{H_{\max}(z)}_{\text{max. phase}}$$

Because zeros are reciprocals,  $H_{\max}(z) = z^{-M_i} H_{\min}(z^{-1})$ ,

where  $M_i$  is the number of zeros of  $H_{\min}(z)$ .

Example:

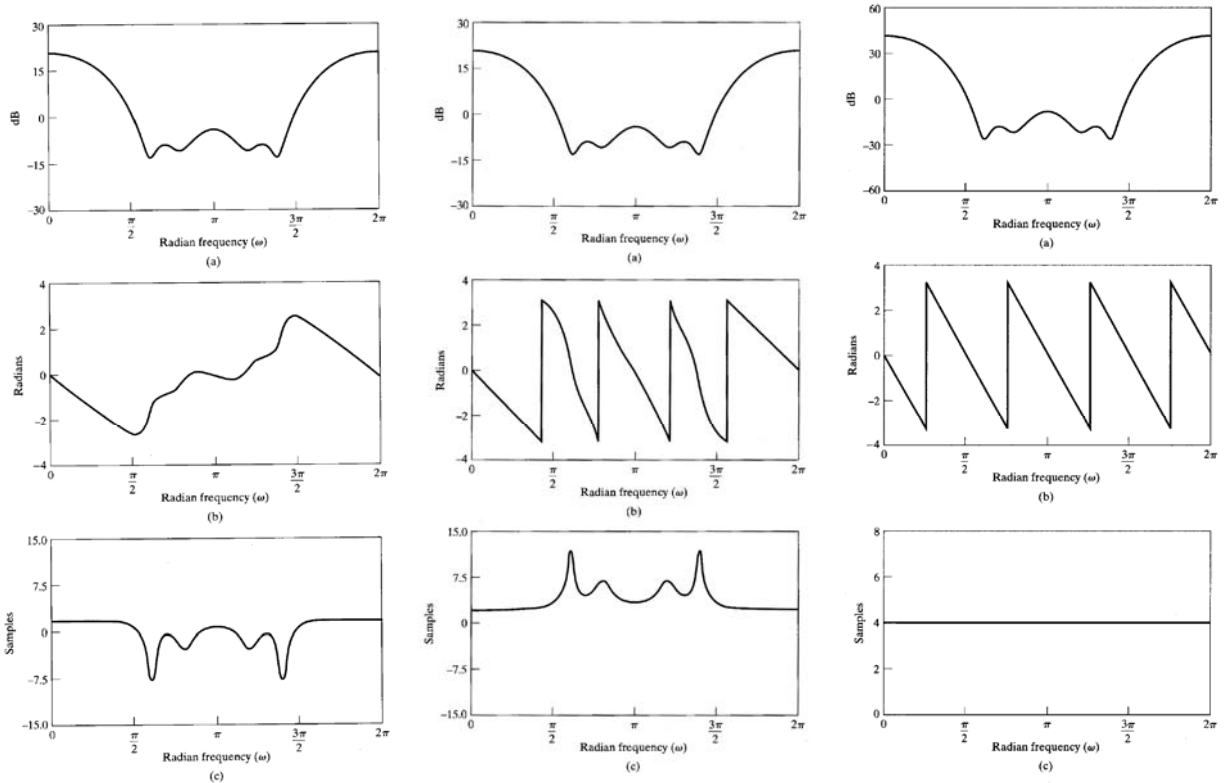
$$H_{\min}(z) = (1.25)^2 (1 - 0.9e^{j0.6\pi} z^{-1})(1 - 0.9e^{-j0.6\pi} z^{-1})(1 - 0.8e^{j0.8\pi} z^{-1})(1 - 0.8e^{-j0.8\pi} z^{-1})$$

$$H_{\max}(z) = (0.9)^2 (1 - 1.1111e^{j0.6\pi} z^{-1})(1 - 1.1111e^{-j0.6\pi} z^{-1})(1 - 1.25e^{j0.8\pi} z^{-1})(1 - 1.25e^{-j0.8\pi} z^{-1})$$

$$H(z) = H_{\min}(z)H_{\max}(z)$$

$$\Rightarrow 20\log_{10}|H(e^{j\omega})| = 20\log_{10}|H_{\min}(e^{j\omega})| + 20\log_{10}|H_{\max}(e^{j\omega})| = 40\log_{10}|H_{\min}(e^{j\omega})|$$

$$\angle H(e^{j\omega}) = \angle H_{\min}(e^{j\omega}) + \angle H_{\max}(e^{j\omega}) = \angle H_{\min}(e^{j\omega}) + (-\omega M_i - \angle H_{\min}(e^{j\omega})) = -\omega M_i$$



**Figure 5.28** Frequency response for minimum-phase system in Example 5.15. (a) Log magnitude. (b) Phase. (c) Group delay. **Figure 5.42** Frequency response of maximum-phase system having the same magnitude as the system in Figure 5.28. (a) Log magnitude. (b) Phase (principal value). (c) Group delay. **Figure 5.43** Frequency response of cascade of maximum-phase and minimum-phase systems, yielding a linear-phase system. (a) Log magnitude. (b) Phase (principal value). (c) Group delay.

$$H_{\min}(z)$$

$$H_{\max}(z)$$

$$H(z)$$

## Type I

$$\begin{aligned} H(e^{j\omega}) &= \sum_{n=0}^M h[n]e^{-j\omega n} \\ &= h[M/2]e^{-j\omega M/2} + \sum_{n=0}^{M/2-1} \{h[n]e^{-j\omega n} + h[M-n]e^{-j\omega(M-n)}\} \\ &= h[M/2]e^{-j\omega M/2} + \sum_{n=0}^{M/2-1} h[n]e^{-j\omega M/2} [e^{-j\omega(n-M/2)} + e^{j\omega(n-M/2)}] \\ &= e^{-j\omega M/2} \{h[M/2] + \sum_{n=0}^{M/2-1} 2h[n]\cos(\omega(n-M/2))\} \end{aligned}$$

■ The delay  $M/2$  is an integer.

$$\blacksquare H(e^{j\omega}) = e^{-j\omega M/2} \sum_{n=0}^{M/2} a[n]\cos(\omega n)$$

$$\textbf{Type II } H(e^{j\omega}) = e^{-j\omega M/2} \sum_{n=1}^{(M+1)/2} b[n]\cos[\omega(n-1/2)]$$

■ The delay  $M/2$  is an integer plus one-half.

$$\textbf{Type III } H(e^{j\omega}) = je^{-j\omega M/2} \sum_{n=1}^{M/2} c[n]\sin(\omega n)$$

■ The delay  $M/2$  is an integer.

$$\textbf{Type IV } H(e^{j\omega}) = je^{-j\omega M/2} \sum_{n=1}^{(M+1)/2} d[n]\sin[\omega(n-1/2)]$$

■ The delay  $M/2$  is an integer plus one-half.